

COMPUTER NETWORK TECHNOLOGY (300)

STATE – 2019

TOTAL POINTS _____ ***(100 points)***

Failure to adhere to any of the following rules will result in disqualification:

- 1. Contestant must hand in this test booklet and all printouts. Failure to do so will result in disqualification.**
- 2. No equipment, supplies, or materials other than those specified for this event are allowed in the testing area. No previous BPA tests and/or sample tests or facsimile (handwritten, photocopied, or keyed) are allowed in the testing area.**
- 3. Electronic devices will be monitored according to ACT standards.**

No more than sixty (60) minutes testing time

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MULTIPLE CHOICE

Identify the letter of the choice that best completes the statement or answers the question.

1. How many broadcast domains are created by a switch by default if the switch has 12 data ports?
 - a. 0
 - b. 1
 - c. 6
 - d. 12
2. What address is a switch most concerned with?
 - a. MAC address
 - b. IP address
 - c. IPv6 address
 - d. port address
3. What address is a router concerned with most?
 - a. source MAC address
 - b. source IP address
 - c. destination IP address
 - d. destination MAC address
4. What are the main groupings termed in DNS?
 - a. Clusters
 - b. Groups
 - c. Sites
 - d. Zones
5. Why might you use port forwarding in your network?
 - a. to control the rate of certain traffic forms
 - b. to direct an external client to an internal resource
 - c. to enable intense logging of traffic
 - d. to prevent certain traffic forms from entering your network
6. What is the *first* part of the four-step DHCP process?
 - a. Acknowledge
 - b. Discover
 - c. Offer
 - d. Request
7. Which of the following would be considered the most legacy of WAN technologies?
 - a. DSL
 - b. dialup
 - c. cable broadband
 - d. ISDN

8. What connector is *commonly* used for asynchronous serial communications with a modem?
 - a. RJ-11
 - b. DB-25
 - c. DB-9
 - d. RJ-45
9. You need to attach the RJ-45 connector to your UTP. What tool would you use?
 - a. wire stripper
 - b. crimper
 - c. punch-down tool
 - d. snips
10. What cable type would you use to connect a router to a switch?
 - a. straight-through
 - b. crossover
 - c. rollover
 - d. plenum
11. What type of network is optimized for the transmission of rich audio and video services?
 - a. WAN
 - b. LAN
 - c. PAN
 - d. Medianet
12. What type of network is a bus star?
 - a. point-to-multipoint
 - b. hybrid
 - c. full mesh
 - d. partial mesh
13. Which of the following is an *advantage* of a peer-to-peer network compared with a client-server network?
 - a. better performance
 - b. less expensive
 - c. simplified administration
 - d. more scalable
14. What is the addressing format that permits the automatic creation of an IPv6 host portion in an address?
 - a. Link local
 - b. EUI-64
 - c. Anycast
 - d. DHCPv6

15. How can the following IPv6 address be condensed?

2009:0123:4040:0000:0000:000A:100B

- a. 2009::123:404:A:100B
 - b. 2009::123:404:A:1B
 - c. 2009:123:4040::A:100B
 - d. 2009::0123:4040::0::000A:100B
16. What is the network address and subnet mask of a default route?
- a. 255.255.255.255/32
 - b. 0.0.0.0/0
 - c. 255.255.255.255/0
 - d. 0.0.0.0/32
17. Which of the following is a distance-vector routing protocol with a maximum usable hop count of 15?
- a. BGP
 - b. IS-IS
 - c. RIP
 - d. OSPF
18. What is the address space reserved for loopback purposes?
- a. 1.0.0.0
 - b. 127.0.0.0
 - c. 172.16.0.0
 - d. 10.10.10.10
19. What device would you choose for an organization that is very concerned about DoS-type attacks against their external router?
- a. content engine
 - b. load balancer
 - c. traffic shaper
 - d. firewall
20. SYSLOG information is often stored in what?
- a. a flat file
 - b. a database
 - c. an unstructured container
 - d. a word document

21. After you've corrected a problem that requires a network reconfiguration, what is a *common* step you must perform?
- a. email the organization
 - b. update your resume
 - c. update the documentation
 - d. phone the CIO
22. Your boss has asked you to gather performance information to provide information on "normal" network utilization. This is referred to as what?
- a. Top Talkers
 - b. SNMP scaping
 - c. Trending
 - d. Baselining
23. Which of the following is the *biggest* concern when dealing with a legacy system?
- a. expense
 - b. expandability
 - c. availability
 - d. security
24. Round-robin is often a technique used in what networking technology?
- a. security segmentation
 - b. load balancing
 - c. testing lab
 - d. guest VLAN
25. Guest networks are often created today using what technology?
- a. fiber
 - b. copper
 - c. wireless
 - d. MMF
26. What is the *safest* option if you are concerned about testing major OS updates in a test lab before deployment?
- a. download updates automatically in off hours
 - b. download updates automatically to a staging server, and then deploy automatically
 - c. download updates at a set time each day
 - d. do not download updates automatically
27. How is VTP information transmitted over the switched network?
- a. using access ports
 - b. using trunk ports
 - c. using EtherChannels
 - d. using SVI interfaces

28. How do you assign an IP address to a VLAN for potential routing on a Layer 3 switch?
- a. use a trunk port
 - b. use an access port
 - c. use an SVI
 - d. this cannot be accomplished
29. You have four links between your switches with no EtherChannel configured; what protocol will dynamically block three of these links?
- a. VTP
 - b. 802.1D
 - c. 802.1Q
 - d. SPAN
30. What device can extend the reach of the wireless LAN?
- a. Layer 2 switch
 - b. Firewall
 - c. Wireless bridge
 - d. IPS
31. What is an important aspect of the Smurf attack?
- a. Spoofing
 - b. Man-in-the-middle
 - c. Reconnaissance
 - d. Packet sniffing
32. What is the term for an attack that was just created?
- a. Code Red
 - b. First Event
 - c. Zero Day
 - d. First Day
33. Refer to the exhibit in the box below. What is the problem with the access list shown?

```
access-list 10 deny host 10.10.10.1  
access-list 10 deny host 192.168.1.100
```

- a. The order of the statements is incorrect.
- b. An access list cannot block two hosts from different subnets.
- c. An access list cannot block specific hosts.
- d. There is no permit statement in the access list.

34. What command would you use on a Linux system to view the IP address on an interface?

- a. ifconfig
- b. ipconfig
- c. view all
- d. netsys

35. What command is similar to **tracert**?

- a. ping
- b. arp
- c. ifconfig
- d. pathping

36. Refer to the exhibit in the box below. What command produced the following results?

```
? (172.16.1.255) at ff:ff:ff:ff:ff:ff on vmnet3 ifscope  
[ethernet]  
? (172.16.2.255) at ff:ff:ff:ff:ff:ff on vmnet4 ifscope  
[ethernet]  
? (172.16.3.255) at ff:ff:ff:ff:ff:ff on vmnet5 ifscope  
[ethernet]  
? (172.16.10.255) at ff:ff:ff:ff:ff:ff on vmnet6 ifscope  
[ethernet]  
? (192.168.0.1) at 90:c7:92:cc:dd:a7 on en1 ifscope [ethernet]  
? (192.168.0.4) at 0:c:29:f9:fc:3f on en1 ifscope [ethernet]  
? (192.168.0.255) at ff:ff:ff:ff:ff:ff on en1 ifscope  
[ethernet]
```

- a. netstat
- b. arp -a
- c. nbtstat
- d. nslookup

37. Which of the following should you ensure if you are adding a new AP to extend the reach of your SSID?

- a. Make sure there is 50 percent overlap
- b. Be sure that the SSID name increments by 1
- c. Be sure to use a non-overlapping channel
- d. Do not use the same security key

38. What is the *best* practice when you are considering the use of new security software?

- a. Deploy the software in stages
- b. Deploy the software in one department only
- c. Deploy the software during normal use hours
- d. Use the software in a test lab.

39. What is the process of integrating multiple streams of information or data at the same time?
- Modulation
 - Multiplexing
 - CSMA/CD
 - CSMA/CA
40. Which port is used by DNS?
- 53
 - 69
 - 23
 - 22
41. What transmission method does 802.11a utilize?
- DSSS
 - OFDM
 - MIMO
 - MU-MIMO
42. Which VoIP and video over IP protocol is used to create media sessions over the network?
- RTP
 - SFTP
 - SIP
 - SSH
43. What transport layer protocol features connectionless communications?
- TCP
 - PIM
 - UDP
 - IGMP
44. What is the max speed for 802.11g?
- 11 Mbps
 - 54 Mbps
 - 100 Mbps
 - 300 Mbps
45. What transmission method is used by 802.11n?
- DSSS
 - OFDM
 - MIMO
 - MU-MIMO

46. What frequency band does 802.11ac use?
- a. 2.4 GHz
 - b. 4 GHz
 - c. 5 GHz
 - d. 10 GHz
47. What is the *maximum* speed for 802.11b?
- a. 11 Mbps
 - b. 54 Mbps
 - c. 100 Mbps
 - d. 300 Mbps
48. Which version of SNMP provides the *highest* level of security?
- a. Version 1
 - b. Version 2
 - c. Version 2c
 - d. Version 3
49. TCP and UDP are found at what layer of the OSI model?
- a. Application layer
 - b. Presentation layer
 - c. Session Layer
 - d. Transport Layer
50. What access method is used in wireless networking today?
- a. CSMA/CD
 - b. VTP
 - c. CSMA/CA
 - d. STP